Dijkstra’s Algorithm on Indian Cities Road Network

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1. Introduction

This project implements Dijkstra’s Algorithm (also known as Uniform Cost Search in AI) to find the shortest path between cities in India based on road distances provided in a CSV file.

2. Algorithm Explanation

Dijkstra’s Algorithm finds the shortest path from a source node to all other nodes in a weighted graph. It works by always expanding the node with the smallest known distance using a priority queue. The algorithm updates distances to neighboring nodes if a shorter path is found.

3. Requirements

- Python 3.x
- CSV file containing city connections
- Required libraries: csv, heapq, collections (default in Python)

4. CSV Format

The CSV file must contain the following columns:
Origin,Destination,Distance

Example:
Hyderabad,Bangalore,570
Bangalore,Chennai,350

5. How to Run the Project

1. Place the CSV file (india_roads.csv) in the same directory as the Python script.
2. Open terminal in the project folder.
3. Run the program using:
   python dijkstra_india.py
4. Enter the start city when prompted.
5. Enter the destination city to view the shortest path.

6. Output

The program displays:
- Shortest distance from the source city to all other cities
- Shortest path to the chosen destination

Example output:

```
Enter start city: Hyderabad

Shortest distances from Hyderabad :

Hyderabad -> Agra = 1822.0 km
Hyderabad -> Delhi = 1582.0 km
Hyderabad -> Lucknow = 2130.0 km
Hyderabad -> Kanpur = 2099.0 km
Hyderabad -> Ahmedabad = 1220.0 km
Hyderabad -> Mumbai = 708.0 km
Hyderabad -> Pune = 560.0 km
Hyderabad -> Jaipur = 1751.0 km
Hyderabad -> Udaipur = 1475.0 km
Hyderabad -> Bengaluru = 576.0 km
Hyderabad -> Hyderabad = 0 km
Hyderabad -> Chennai = 626.0 km
Hyderabad -> Goa = 674.0 km
Hyderabad -> Bhubaneswar = 1059.0 km
Hyderabad -> Kolkata = 1489.0 km
Hyderabad -> Vishakhapatnam = 618.0 km
Hyderabad -> Patna = 1890.0 km
Hyderabad -> Kochi = 1123.0 km
Hyderabad -> Thiruvananthapuram = 1329.0 km
Hyderabad -> Varanasi = 2146.0 km

Enter destination city to see path: Varanasi

Shortest path:
Hyderabad -> Vishakhapatnam -> Bhubaneswar -> Patna -> Varanasi
```

7. Conclusion

This project demonstrates how graph search algorithms like Dijkstra's Algorithm can be applied to real-world problems such as route optimization and navigation systems.